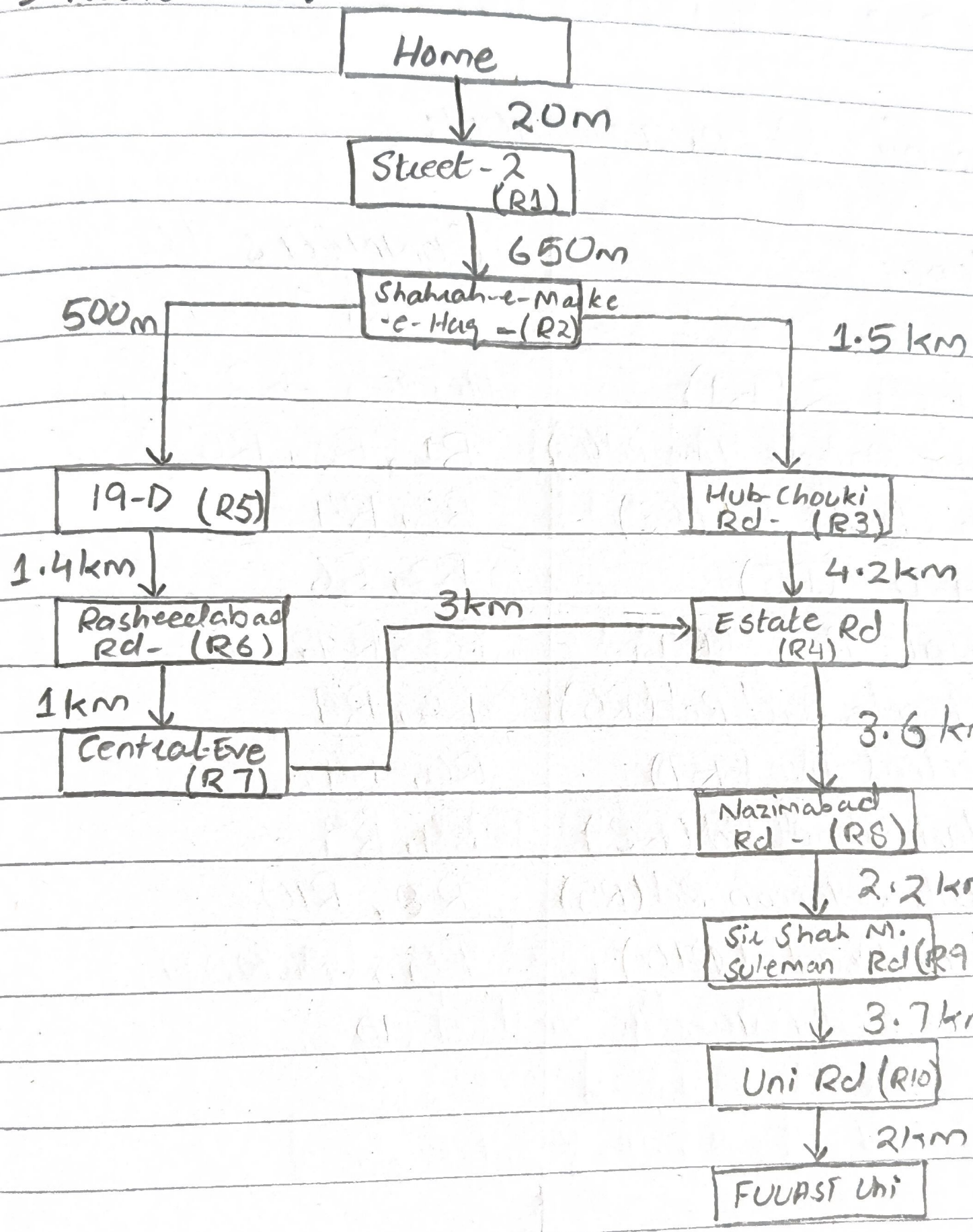
ARISHA - 28122106 - 6A

HOME TO UNI ROUTE

BFS

1. ROUTE MAP:



Route 1:

0.02 + 0.65 + 1.5 + 4.2 + 3.6 + 2.2 + 3.7 + 2 = 17.87 km

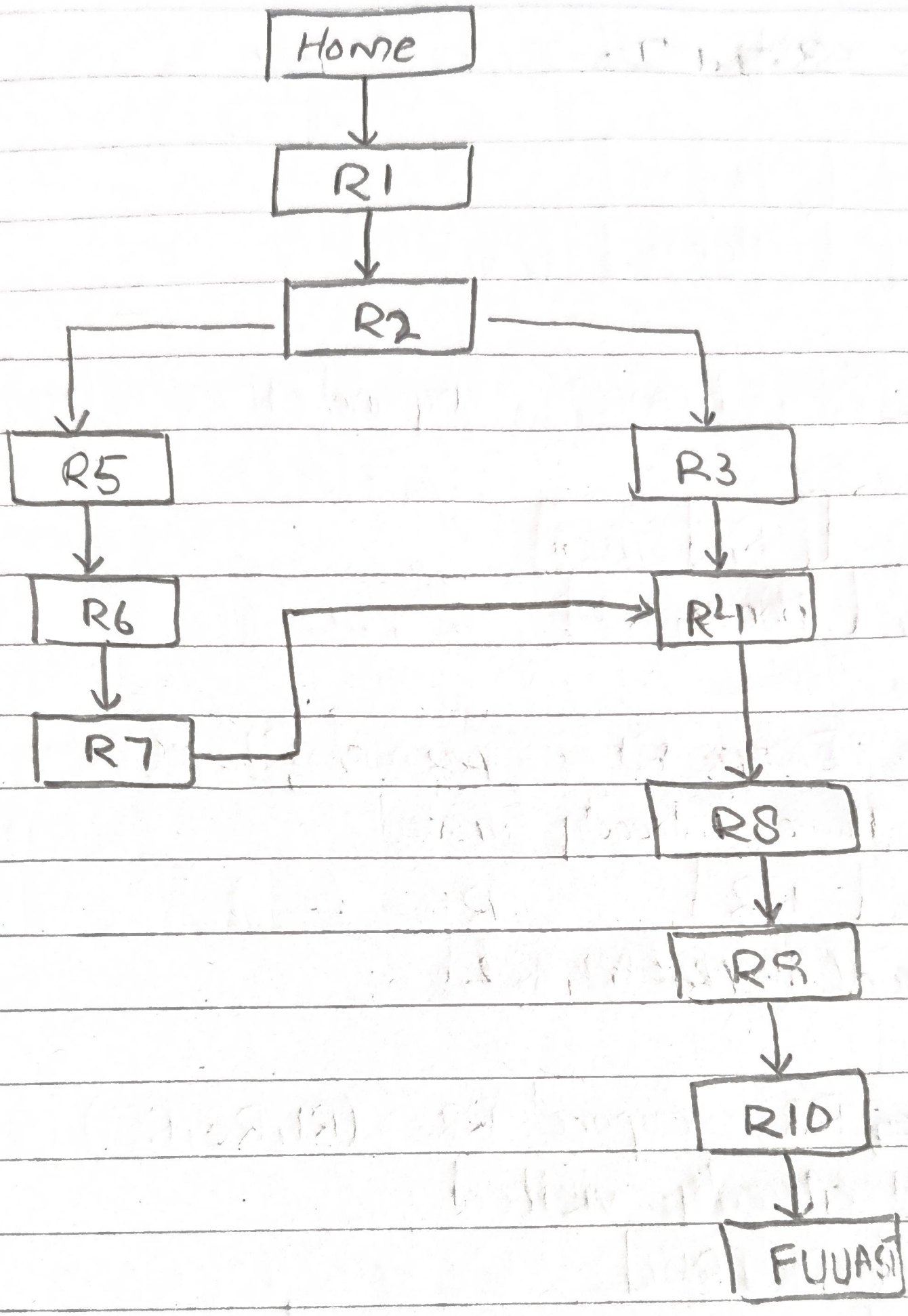
Route 2:

0.2 + 0.65 + 0.5 + 1.4 + 1 + 3 + 3.6 + 2.2 + 3.7 + 2 = 18.25 KM

GRAPH REPRESENTATION:

NODE	CONNECTS TO
Home	R1
Street 2 (R1)	Home, R2
Shahrah-e-Mosha-e-Haq Rd (R2)	R1, R3, R5
Hub Chouki Rd (R3)	R2, R4
19-D (R5)	R2, R6
Estate Ave (R4)	R3, R7, R8
Rasheeda bad Rd (R6)	R5, R7
Centoal Ave (R7)	R6, R4
Nazimabad Rd (R8)	R4, R9
Sir Shah Suleman Rd (R9)	R8, R10
Uni Road (R10)	R9, FUUAST
FUUAST	R10

BFS Tree:



BFS Trace:

Step 0: Initialization

Q = [Home]
V = [Home]

Step 1: Deq Home Expand Home → (R1)

Q = [R1]
V = [Home, R1]

Step 2: Deq R1 Expand R1 → (Home, R2)
Home already visited

Q = [R2]
V = [Home, R1, R2]

Step 3: Deq R2 expand R2 → (R1, R3, R5)
R1 already visited

Q = [R3, R5]
V = ['', '', R2, R3, R5]

Step 4: Deq R3 expand R3 → (R2, R4)
R2 already visited

Q = [R5, R4]

V= [- - - - - - , R3, R5, R4]

Step 5: Deq R5 expand R5 → (R2, R6)
R2 already visited

Q= [R4, R6]

V= [- - - - - - , R5, R4, R6]

Step 6: Deq R4 expand R4 → (R3, R7, R8)
R3 (AV)

Q= [R6, R7, R8]

V= [- - - - - , R6, R7, R8]

Step 7: Deq R6 expand R6 → (R5, R7)
both AV

Q= [R7, R8]

V= [- - - - - , R6, R7, R8]

Step 8: Deq R7 expand R7 → (R6, R4)
both AV

Q= [R8]

V= [- - - - - , R7, R8]

Step 9: Deq R8 expand R8 → (R4, R9)
R4 (AV)

Q= [R9]

V = [- - - - , R8, R9]

Step 10: Deq R9 expand R9 → (R8, R10)
R8(AV)

Q = [R10]
V = [- - - - , R9, R10]

Step 11: Deq R10 expand R10 → (R9, FUUA
R9(AV)

Q = [FUUAST]
V = [- - - - , R9, R10, FUUASI]

Step 12: Deq FUUASI expand FUUASI
→ (R10)
R10 (AV)

Goal Achieved.

Path:

→ Shortest Path:

Home → R1 → R2 → R3 → R4 → R8 →
R9 → R10 → FUUASI
= 8 edges.

→ Alternate Path:

Home → R1 → R2 → R5 → R6 → R7 →
R4 → R8 → R9 → R10 → FUUAST

= 10 edges